



CSE 598 008 EECS 498 003 Lab 1

Tianyiming (Henry) Jing
September 3, 2026

Based on slides by Ivan Wei (Fall 2025).

About Me

- Name: 井 田 一 鸣
Jing Tian Yi Ming (Jing-tyen Ee-ming)
Henry
- Interests: Logic, ACG (Anime, Comic, and Games), Music, Movies

Office Hours

- Location: 1270A Leinweber
- Time: Monday, 11:00 – 12:00
Wednesday, 12:00 – 13:00

Lab Content

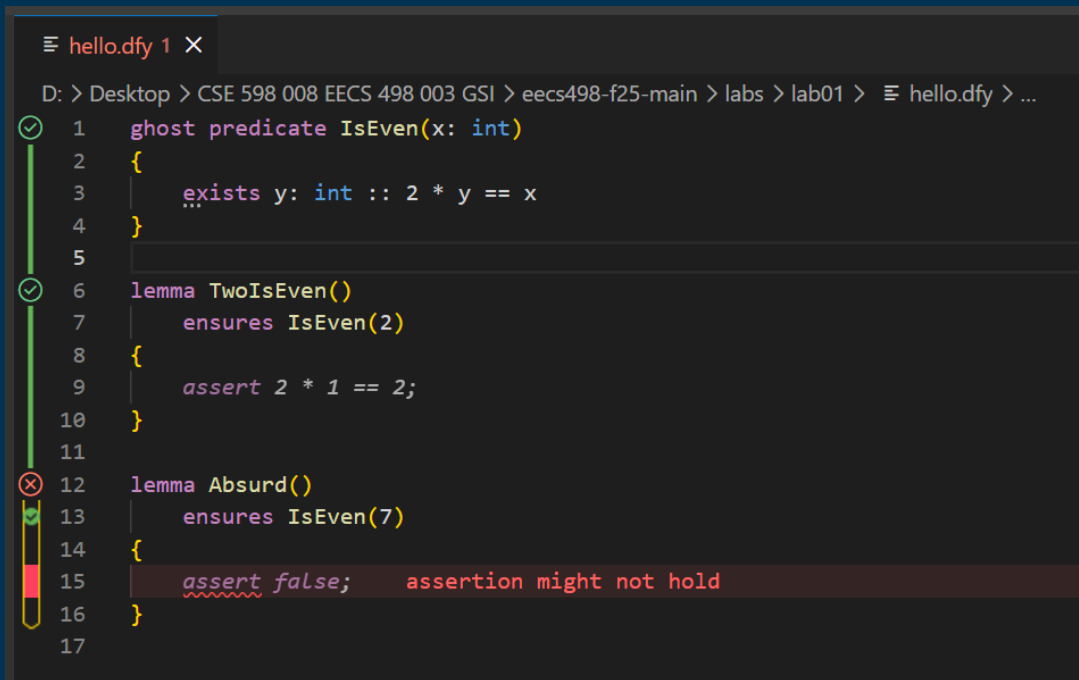
- Labs will mostly be:
 - interactive exercises and practice
 - “office hours”

GitHub Repository

- <https://github.com/GLaDOS-Michigan/eecs498-f26.git>
- `git clone git@github.com:GLaDOS-Michigan/eecs498-f26.git`

Dafny

VSCode Extension



```
hello.dfy 1 X
D: > Desktop > CSE 598 008 EECS 498 003 GSI > eecs498-f25-main > labs > lab01 > hello.dfy > ...
1  ghost predicate IsEven(x: int)
2  {
3      exists y: int :: 2 * y == x
4  }
5
6  lemma TwoIsEven()
7      ensures IsEven(2)
8  {
9      assert 2 * 1 == 2;
10 }
11
12 lemma Absurd()
13     ensures IsEven(7)
14 {
15     assert false;    assertion might not hold
16 }
17
```

CLI Tool

```
PS D:\Desktop\CSE 598 008 EECS 498 003 GSI\eeecs498-f26\labs\lab01> dafny verify hello.dfy
hello.dfy(15,4): Error: assertion might not hold

15 |         assert false;
    |         ^^^^^^^

Dafny program verifier finished with 1 verified, 1 error
PS D:\Desktop\CSE 598 008 EECS 498 003 GSI\eeecs498-f26\labs\lab01> |
```

Installation

- Prerequisite: .NET SDK

- Instructions

- <https://learn.microsoft.com/en-us/dotnet/core/install/>

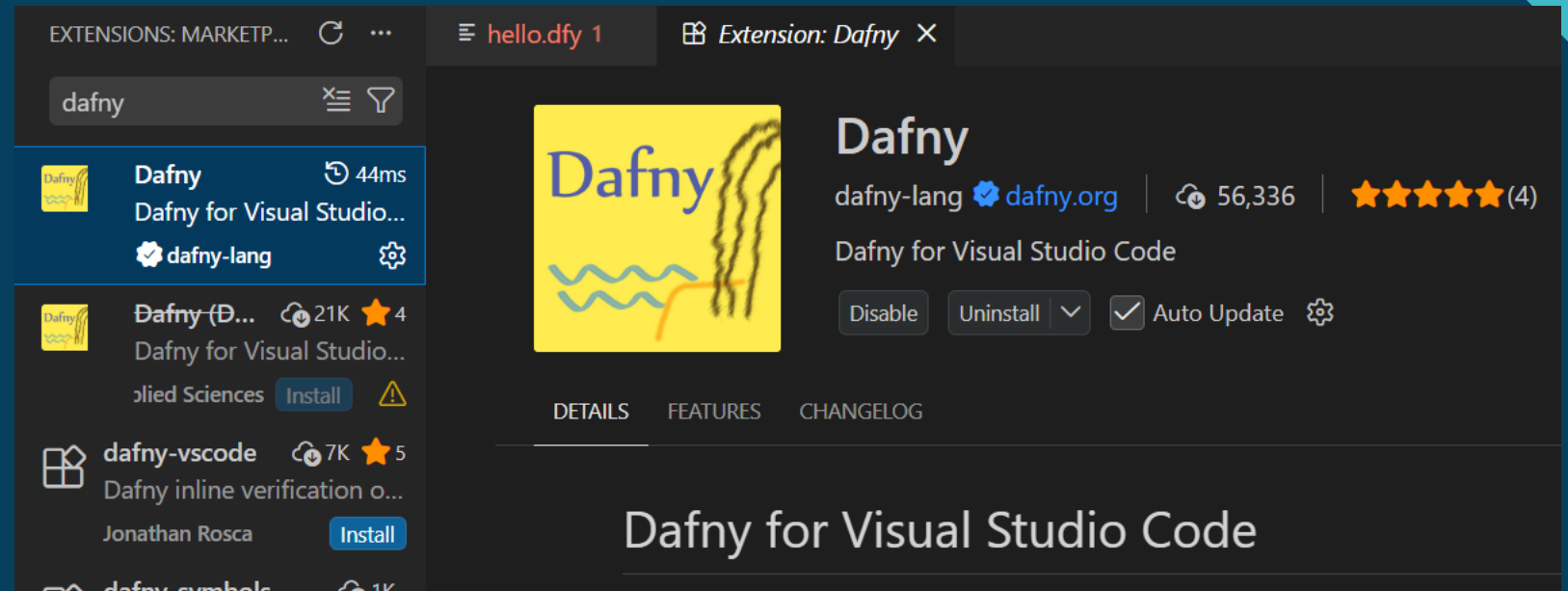
- Verify .NET is installed

- `dotnet --version`

Installation

- Dafny 4.11.0 (current latest version)
 - Instructions
<https://dafny.org/latest/Installation>
 - VSCode extension: from marketplace
 - If installing binary, to verify Dafny is installed

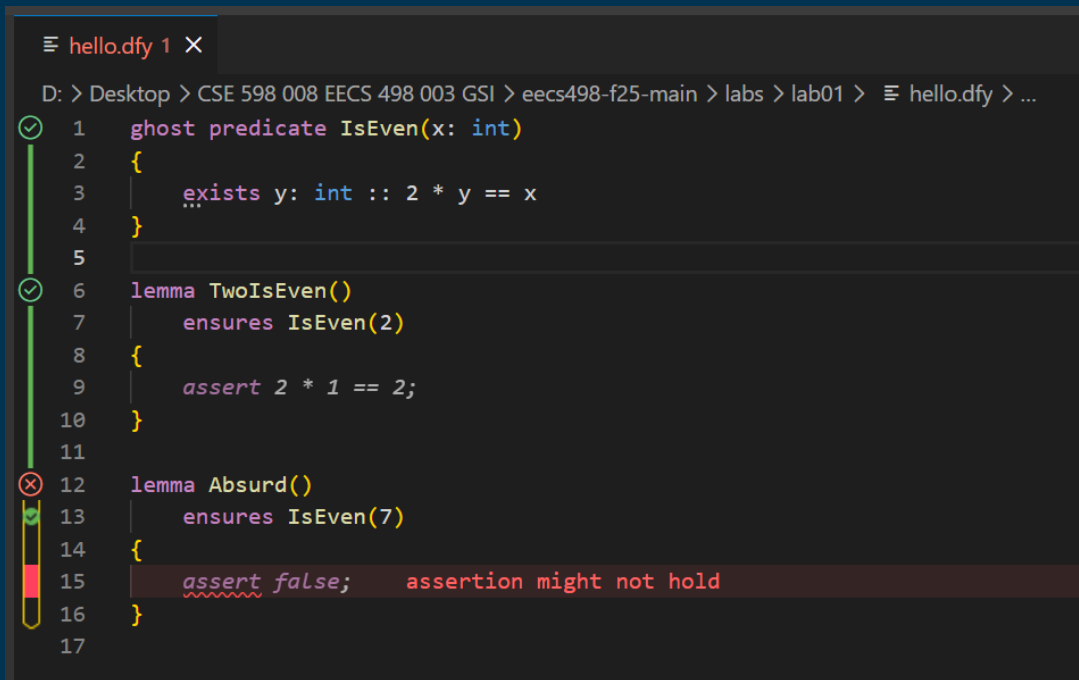
`dafny --version`



Testing

Get hello.dfy from the GitHub repository

VSCode Extension



```
1 ghost predicate IsEven(x: int)
2 {
3     exists y: int :: 2 * y == x
4 }
5
6 lemma TwoIsEven()
7     ensures IsEven(2)
8 {
9     assert 2 * 1 == 2;
10 }
11
12 lemma Absurd()
13     ensures IsEven(7)
14 {
15     assert false;    assertion might not hold
16 }
17
```

CLI Tool

```
PS D:\Desktop\CSE 598 008 EECS 498 003 GSI\eecs498-f26\labs\lab01> dafny verify hello.dfy
hello.dfy(15,4): Error: assertion might not hold
```

```
15 |         assert false;
    |         ^^^^^^^
```

```
Dafny program verifier finished with 1 verified, 1 error
```

```
PS D:\Desktop\CSE 598 008 EECS 498 003 GSI\eecs498-f26\labs\lab01> |
```

VSCode: onChange VS onSave

- Open VsCode settings (Ctrl/Cmd + ,)
- Search “dafny automatic verification”
 - OnChange: runs everytime you edit (default)
 - OnSave: runs everytime you save

Troubleshooting

- The extension complains about .NET even after install.
 - Clear the setting Dafny: Dotnet Executable Path in VSCode settings.
 - Make sure your dotnet installation is in your PATH.
- The extension attempts to auto-download the Dafny binary and fails.
 - Download the binary yourself
 - Set the VSCode setting “Dafny: Version” to custom, then set the VSCode setting “Dafny: Cli Path” to the path of the Dafny executable file.

Extra

<i>Rule of inference</i>	<i>Name</i>	<i>Rule of inference</i>	<i>Name</i>
$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$	Modus Tollens	$\frac{p}{\therefore p \vee q}$	Disjunctive Amplification
$\frac{p \rightarrow q \quad p}{\therefore q}$	Modus Ponens	$\frac{\neg p \rightarrow \text{False}}{\therefore p}$	Contradiction
$\frac{p \wedge q}{\therefore p}$	Simplification	$\frac{p \wedge q \quad p \rightarrow (q \rightarrow r)}{\therefore r}$	Conditional Proof
$\frac{p \quad q}{\therefore p \wedge q}$	Conjunction	$\frac{p \rightarrow r \quad q \rightarrow r}{\therefore (p \vee q) \rightarrow r}$	Proof by Cases
$\frac{p \vee q \quad \neg p}{\therefore q}$	Disjunctive Syllogism	$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	Law of Syllogism

https://www.researchgate.net/figure/Major-standard-inference-rules-for-predicate-logic_fig6_348286706

Extra

Name	Rule	Example
Universal Instantiation	$\forall xP(x)$ $\therefore P(c)$	"All women are brave." "Therefore, Lily is brave."
Universal Generalization	$P(c)$ for an arbitrary c $\therefore \forall xP(x)$	"Lily is brave." "Therefore, all women are brave."
Existential Instantiation	$\exists xP(x)$ $\therefore P(c)$ for some element c	"There is someone who ran a mile in 4 minutes." "Let's call him Sparky and say that Sparky ran a mile in 4 minutes."
Existential Generalization	$P(c)$ for some element c $\therefore \exists xP(x)$	"Sparky ran a mile in 4 minutes." "Therefore, someone ran a mile in 4 minutes."